

ATVB IN FOCUS: Sleep and Cardiovascular Disease

Series Editors: Phyllis Zee and Michael Grandner

Sleep, Neural Circulatory Control, and Cardiovascular Disease: A Mechanistic Review

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ABSTRACT: Sleep is an active period of profound autonomic fluctuation, cycling between the parasympathetic dominance of nonrapid eye movement sleep and the sympathetic/parasympathetic volatility of rapid eye movement sleep. Sleep-disordered breathing, a spectrum of disorders marked by recurrent ventilatory instability and intermittent hypoxia during sleep, particularly obstructive sleep apnea, pathologically amplifies this volatility, transforming sleep into a nightly cascade of severe autonomic and hemodynamic stress. The cardinal features of sleep-disordered breathing, intermittent hypoxia, recurrent arousals, and marked intrathoracic pressure swings, act synergistically to drive chronic, 24-hour sympathetic overactivity, chemoreflex sensitization, and maladaptive neuroplasticity. These effects are mediated at a cellular level by oxidative stress, systemic inflammation, endothelial dysfunction, and neuroendocrine dysregulation. This persistent autonomic reset provides a direct mechanistic link to cardiovascular consequences. It is likely a primary driver of hypertension, blunting the nocturnal blood pressure dip and promoting sustained 24-hour sympathoexcitation. It fosters a proarrhythmic substrate for atrial fibrillation through mechanical stress, which drives atrial remodeling and autonomic conflict. Furthermore, sleep-disordered breathing contributes to myocardial ischemia by increasing myocardial oxygen demand and promoting a prothrombotic state. Beyond chronic disease, sleep-related autonomic shifts can act as acute triggers for malignant arrhythmias in individuals with vulnerable substrates, such as inherited channelopathies, a risk that may be significantly amplified by comorbid sleep-disordered breathing. This review delineates the critical neural and cellular pathways connecting sleep, autonomic dysregulation, and cardiovascular risk.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: autonomic nervous system ■ cardiovascular diseases ■ memory consolidation ■ prevalence ■ sleep apnea syndromes ■ sleep apnea, obstructive ■ wakefulness

Sleep is fundamental to life and critical to preserving health and well-being. Despite its importance for neuronal development, metabolism, cognitive function, and memory consolidation, the precise underlying mechanisms that make sleep essential for survival remain one of the most significant unanswered questions in human biology.^{1–5} The cyclical nature of sleep and wakefulness suggests a regular alternation between distinct neural systems, each working to maintain 1 of the 2 states and governed by a delicate interdependent balance of neurotransmitters.⁶ Both sleep quality and sleep quantity are critical to the normal operation of nearly every system in the body, and derangements in either lead to demonstrable consequences.^{7–9}

The spectrum of sleep disorders is vast, encompassing nearly 90 distinct conditions, and their prevalence is

increasingly recognized.^{7,10} The systemic, downstream consequences of these disorders are also being progressively determined, firmly linking them to increased cardiovascular morbidity and mortality.^{11–13} By 2030, 1 in 5 individuals in the United States is projected to be over the age of 65 years.¹⁴ Aging is associated with physiological, though not always pathological, changes in sleep architecture, which increase susceptibility to sleep disturbances.¹⁵ Between 2000 and 2010, the prevalence of diagnosed sleep disorders in the United States rose nearly 6-fold,¹⁶ and estimates suggest that between 17% and 56% of the general population experience

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Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

ACTH	adrenocorticotrophic hormone
AF	atrial fibrillation
AHI	apnea-hypopnea index
BNP	B-type natriuretic peptide
CIH	chronic intermittent hypoxia
COX	cyclooxygenase
CPAP	continuous positive airway pressure
CRP	C-reactive protein
CSA	central sleep apnea
eNOS	endothelial nitric oxide synthase
EPO	erythropoietin
ET-1	endothelin-1
HIF	hypoxia-inducible factor
hs-CRP	high-sensitivity C-reactive protein
ICAM-1	intercellular adhesion molecule-1
IDO-1	indoleamine 2,3-dioxygenase-1
IH	intermittent hypoxia
IL	interleukin
LQTS	long-QT syndrome
MAO-A	monoamine oxidase A
MCH	melanin-concentrating hormone
NF-κB	nuclear factor-κB
NLRP3	NLR family pyrin domain-containing 3
Nox2	NADPH oxidase 2
NREM	nonrapid eye movement
OSA	obstructive sleep apnea
PHD	prolyl hydroxylase domain
REM	rapid eye movement
ROS	reactive oxygen species
RVOT-VT	right ventricular outflow tract ventricular tachycardia
SDB	sleep-disordered breathing
Sod2	superoxide dismutase 2
TNF-α	tumor necrosis factor-α
TrkB	tropomyosin receptor kinase B
VCAM-1	vascular cell adhesion molecule-1
VEGF	vascular endothelial growth factor
VHL	von Hippel-Lindau
VLPO	ventrolateral preoptic area
VT	ventricular tachycardia

some form of sleep disorder.^{17,18} A critical pathway connecting sleep disruption to these adverse outcomes, particularly cardiovascular disease, is through dysregulation of the autonomic nervous system. Patients with sleep disorders frequently report symptoms suggestive of autonomic nervous system dysfunction, and conversely, individuals with autonomic nervous system abnormalities often experience some form of sleep disturbance.¹⁹

Highlights

- Normal sleep is a period of active autonomic fluctuation, and the cyclical shift between parasympathetic-dominant nonrapid eye movement sleep and sympathetic-volatile rapid eye movement sleep creates a baseline of cardiovascular vulnerability.
- Nonrespiratory sleep disturbances, including insomnia and behavioral sleep deprivation, drive chronic 24-hour sympathetic hyperarousal, blunt normal nocturnal blood pressure dipping, and trigger systemic inflammation.
- Sleep-disordered breathing pathologically amplifies sleep-related autonomic lability, as intermittent hypoxia and intrathoracic pressure swings induce chemoreflex sensitization and maladaptive neuroplasticity.
- These persistent autonomic and cellular disruptions provide a direct mechanistic bridge linking the full spectrum of sleep dysregulation to chronic cardiovascular conditions, including hypertension, heart failure, and atrial fibrillation.
- The profound autonomic shifts occurring during sleep can act as acute triggers for life-threatening arrhythmias in patients with preexisting vulnerable substrates, underscoring the critical need to identify early proximal clinical end points before irreversible injury occurs.

In this review, we will explore the complex, bidirectional relationship between sleep, autonomic neural circulatory regulation, and cardiovascular disease risk, examining the interplay in both health and disease.

THE NEUROBIOLOGY OF NORMAL SLEEP

For sleep to occur, wakefulness must first be overcome. Wakefulness is maintained through cortical activation by the ascending arousal system that is directed by an extensive network of subcortical pathways and excitatory neurotransmitters working interdependently. The arousal system facilitates the consolidation of sleep into a single, continuous period. Furthermore, it assists in the entrainment of the sleep-wake cycle to the external light-dark cycle.²⁰ The brainstem directs wakefulness through 2 methods. First, the laterodorsal tegmentum (pons) and pedunculopontine tegmentum release acetylcholine in the greatest amounts during waking and rapid eye movement (REM) sleep but in minimal levels during nonrapid eye movement (NREM) sleep.²¹ This cholinergic activity serves to activate thalamic areas, thereby maintaining sensory transmission to the cortex. The second pathway involves control over hypothalamic areas, driven by 4 neurotransmitters: norepinephrine from the locus coeruleus, dopamine from ventral periaqueductal gray matter, serotonin from the

dorsal raphe nucleus (the largest group of serotonin-producing cells in the brain), and histamine from the tuberomammillary nucleus.^{22–24} These neurotransmitters converge on the perifornical area, where they interact with a network of orexin-containing, cholinergic (acetylcholine), and GABAergic neurons to broadly activate the cortex and stabilize wakefulness. The transition into sleep is further shaped by NREM-promoting inhibitory systems in the ventrolateral preoptic area (VLPO) and medullary parafacial zone, while REM sleep is generated by a pontine REM-on switch centered in the subcoeruleus/sublaterodorsal region, with cholinergic activation emerging as monoaminergic drive declines (Figure 1A).^{25–28}

While the precise trigger for the initiation of sleep remains unclear, the primary theory involves the homeostatic accumulation of adenosine in the basal forebrain throughout the day.²⁹ This adenosine buildup serves to activate inhibitory neurons within the VLPO, which initiates and maintains sleep through suppression of the ascending arousal system.^{30,31} In addition, the entire mechanism is modulated by the central circadian system (suprachiasmatic nucleus), which provides input to the VLPO to align the sleep-wake cycle with the day-night cycle.³⁰ During sleep, cholinergic neurons actively promote REM sleep while simultaneously inhibiting the noradrenergic neurons of the locus coeruleus and the serotonergic neurons of the raphe nucleus.³² This

reciprocal activity between the cholinergic (REM-on) and aminergic (REM-off) systems drives the characteristic cycling between NREM sleep and REM sleep.³³ This REM-state organization is consistent with established models in which REM generation depends on a brainstem REM-on switch, particularly the subcoeruleus/sublaterodorsal complex, rather than on a simple 1-way inhibitory action of cholinergic neurons on aminergic nuclei.^{34,35} MCH (melanin-concentrating hormone) neurons may contribute to REM sleep regulation and interact with REM-related arousal circuits, potentially through modulation of REM-related network balance (Figure 1B).^{36–38}

A standard sleep cycle is composed of 2 distinct phases, defined by electroencephalographic criteria: NREM and REM sleep. The NREM phase is further divided into 3 stages: N1, N2, and N3, representing progressively deeper levels of sleep. This progression is characterized by a gradual shift in the electroencephalogram, with wave frequency decreasing as amplitude increases.^{24,39} Roughly 75% of total sleep occurs during NREM, with the majority of this time spent in stage N2.⁴⁰ After NREM, sleep transitions into the REM stage, completing the cycle. A typical progression follows the sequence of N1, N2, N3, a return to N2, and then REM.⁴¹ REM sleep itself consists of 2 subtypes: tonic REM, characterized by generalized muscle atonia, and phasic REM, which is driven by sympathetic activation

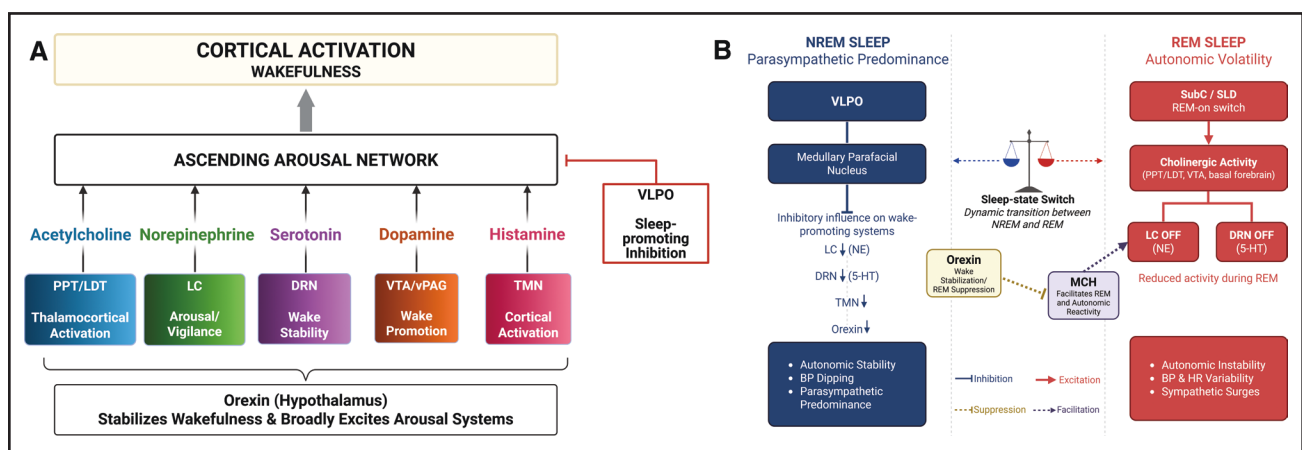


Figure 1. Simplified schematics illustrating the neural circuits underlying wakefulness and sleep-state transitions.

A, Simplified schematic of the neurobiology of wakefulness. Wakefulness is sustained by parallel ascending arousal pathways that converge on cortical activation. Orexin from the hypothalamus stabilizes wakefulness and broadly excites arousal systems, including the pedunculoventral/rostral tegmental nuclei (PPT/LDT; acetylcholine), locus coeruleus (LC; norepinephrine [NE]), dorsal raphe nucleus (DRN; serotonin [5-HT]), ventral periaqueductal gray (vPAG)/ventral tegmental area (VTA; dopamine), and tuberomammillary nucleus (TMN; histamine). The ventrolateral preoptic area (VLPO) provides sleep-promoting inhibition. (Copyright Mayo Foundation, used with permission.) **B**, Simplified schematic of the neural circuitry regulating sleep-state transitions. Sleep-state regulation reflects dynamic interactions between nonrapid eye movement (NREM)–promoting and rapid eye movement (REM)–promoting neural systems. NREM sleep is supported by the VLPO and medullary parafacial nucleus, which inhibit wake-promoting systems and promote parasympathetic predominance and blood pressure (BP) dipping. REM sleep is generated by a pontine REM-on switch centered in the subcoeruleus/sublaterodorsal region (SubC/SLD), which contains excitatory glutamatergic neurons that promote rapid eye movements, theta-frequency EEG oscillations, and REM sleep atonia. This state is accompanied by increased cholinergic activity in the PPT and LDT, along with reduced locus coeruleus (LC; norepinephrine, NE) and dorsal raphe nucleus (DRN; serotonin, 5-HT) activity, contributing to autonomic volatility, sympathetic surges, and increased blood pressure and heart rate (HR) variability. Figure 1A created in BioRender. KARIM, S. (2026) <https://BioRender.com/ouyvwth>. Figure 1B created in BioRender. KARIM, S. (2026) <https://BioRender.com/mqo47r8>.

and features bursts of REMs along with fluctuations in cardiovascular and respiratory activity, often associated with dreaming.^{42–44} Each cycle lasts ≈ 90 to 110 minutes, and a typical night of rest includes 4 to 5 such cycles.²⁴

THE AUTONOMIC PROFILE OF NORMAL SLEEP

There are several profound physiological and autonomic differences between NREM sleep and REM sleep that are fundamental to understanding nocturnal cardiovascular risk. Where NREM sleep represents a state of parasympathetic dominance and autonomic stability, REM sleep is characterized by sympathetic and parasympathetic activation, volatility, and instability.^{45–47} While this REM-related autonomic volatility is normal, it creates a period of inherent cardiovascular vulnerability, which may be exacerbated by underlying substrate pathology and may be dramatically potentiated by amplifiers, such as sleep-disordered breathing (SDB).^{48,49}

During NREM sleep, parasympathetic activity prevails, leading to the normal nocturnal dipping phenomenon, where mean arterial pressure is typically 10% to 15% lower than during wakefulness and heart rate slows.⁵⁰ NREM-promoting neurons within the VLPO also inhibit hypothalamic areas responsible for sympathoexcitatory and pressor functions, in turn suppressing the release of the potent vasoconstrictor vasopressin.⁵¹ This central control, combined with a highly responsive chemoreflex system, ensures cardiorespiratory stability during NREM sleep.⁵² This stability degrades, however, around the transition from NREM to REM sleep.⁴⁶ The associated increased fluctuation in blood pressure may induce baroreflex instability, a condition defined as increased loop gain, which represents an unstable, over-responsive cardiorespiratory control system.⁵³ In addition to VLPO-mediated sleep promotion, medullary parafacial neurons contribute to NREM sleep generation and the associated autonomic stability.⁵⁴

During REM sleep, sympathetic nervous system activity is characterized by marked fluctuations and overall increased activity compared with NREM sleep.⁵⁵ Micro-neurography readings of muscle sympathetic nerve activity recorded with electrode placement in the common peroneal nerve in healthy adults demonstrate that muscle sympathetic nerve activity decreases progressively with deeper NREM sleep stages, reflecting central suppression of sympathetic outflow. In contrast, during REM sleep, sympathetic nerve activity increases significantly, often surging to levels over 200% higher than during wakefulness, which is associated with pronounced variability in heart rate and arterial blood pressure, reflecting autonomic instability.^{46,55,56} This heightened sympathetic activity is also confirmed by biochemical markers: urinary catecholamine excretion (epinephrine and norepinephrine metabolites) positively correlates with the proportion

of REM sleep.⁵⁷ In addition, elevated catecholamine levels are associated with concentration-dependent platelet activation.⁵⁸ While REM sleep itself is not directly prothrombotic, sleep disorders such as obstructive sleep apnea (OSA) that predominate during REM may drive further platelet activation and a heightened likelihood of thrombosis through increased hypoxia, sympathetic activation, and inflammation. These consistent findings across both neural and biochemical modalities firmly link REM sleep to a state of heightened sympathetic drive (Figure 2).

In certain high-risk populations, the increased sympathetic activity and cardiovascular instability observed during REM sleep may precipitate adverse cardiovascular events. Observational data suggest that the threshold for myocardial ischemia may be lower at night, and strong associations have been demonstrated between episodes of nocturnal angina and REM sleep in patients with severe coronary artery disease.^{48,59} During REM sleep, surges in heart rate and blood pressure elevate myocardial oxygen demand, which may exceed supply in susceptible patients and lead to ischemic episodes.⁶⁰ This is particularly evident in patients with OSA, who demonstrated dynamic ST-segment depression during periods of REM sleep, where ischemic activity was tightly coupled with periods of high apneic activity and significant hypoxemia.⁶¹ Furthermore, treatment with continuous positive airway pressure (CPAP) therapy has been associated with reduced duration of apnea-hypopnea-related ST-segment depression (Figure 3).^{62–64}

While arrhythmias occur more frequently during REM sleep, arrhythmias that are directly attributable to REM alone are rare. Although REM sleep is characterized by overall increased sympathetic activity, in patients with REM-related arrhythmias, there may be transient shifts toward vagal dominance, particularly during bursts of REMs.^{65,66} Heart rate variability analyses performed in patients with REM sleep-related arrhythmias have demonstrated uniquely blunted shifts to the sympathetic activation typically observed upon the change from slow-wave (NREM) to REM sleep.⁶⁶ The underlying pathophysiology likely involves abnormal central autonomic neural-cardiac inputs, with possible anomalous baroreflex or parasympathetic surges during REM sleep, leading to episodes of sinus arrest or atrioventricular block.^{65,67} Treatment is individualized, based on the presence of symptoms or high-risk conduction abnormalities, such as complete atrioventricular block. These events are most commonly described in otherwise healthy individuals with normal conduction system function, in whom treatment may not be needed. Importantly, this syndrome is distinct from arrhythmia related to SDB or underlying structural heart disease.^{65,66,68} REM sleep-related bradyarrhythmia syndrome is rare, but documented cases have demonstrated sinus pauses up to 12.5 seconds and atrioventricular block during REM sleep, with normal

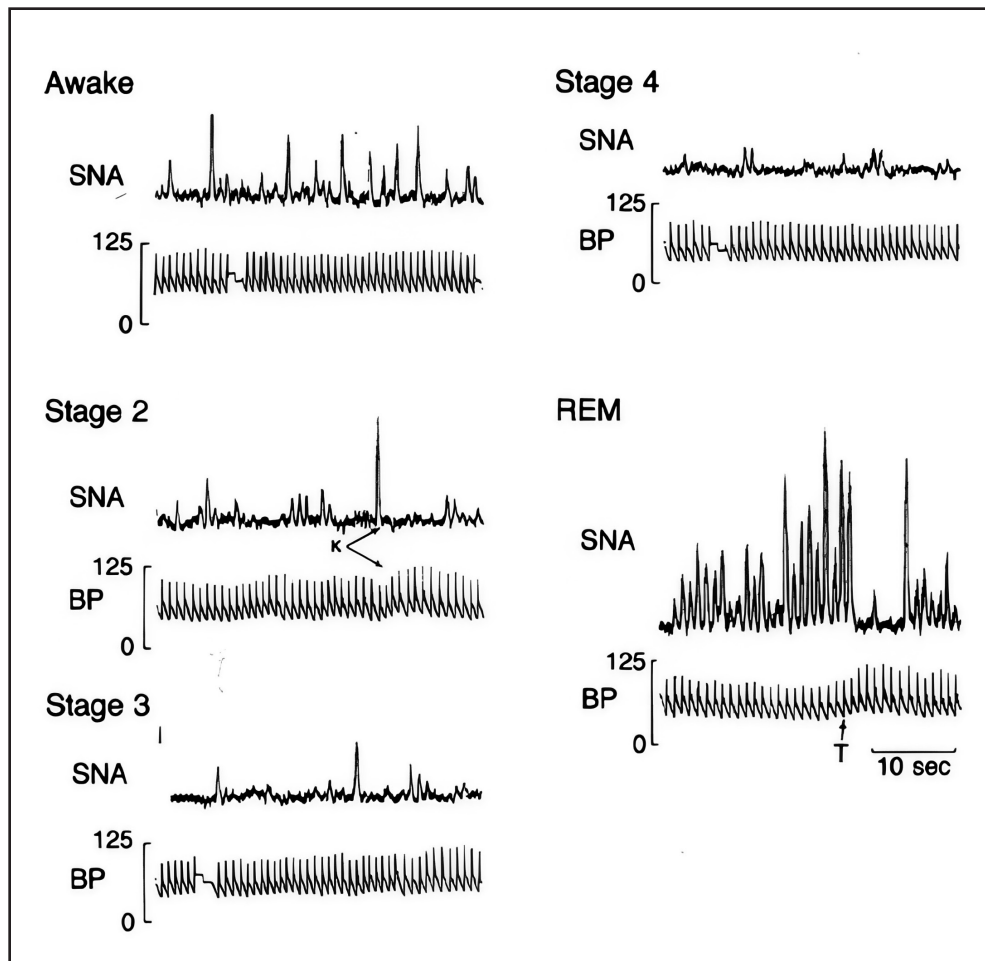


Figure 2. The autonomic profile of normal sleep.

Microneurography recordings illustrating muscle sympathetic nerve activity (SNA) and blood pressure (BP) in a healthy subject during wakefulness and different sleep stages. During nonrapid eye movement sleep, there is a progressive decrease in both SNA and BP from stage 2 to stage 4, reflecting increased parasympathetic dominance and autonomic stability. In sharp contrast, rapid eye movement (REM) sleep is characterized by marked autonomic volatility, with frequent, high-amplitude bursts of SNA and significant fluctuations in BP, often exceeding waking levels. (K=K-complex; T=muscle twitch). Recreated with permission.⁴⁶

cardiac rhythm during wakefulness and NREM sleep.⁶⁸ This arrhythmogenic potential of vagal tone is not limited to rare bradyarrhythmias. While the development of atrial fibrillation (AF) is complex and not limited to changes in a single autonomic component, increased vagal tone has also been found to trigger AF.^{69,70} Given that REM sleep is associated with heightened sympathetic activity, it has been suggested that reduced REM duration exposes the heart to greater vagal predominance, thereby potentially increasing the risk of developing AF.⁷¹

CARDIOVASCULAR IMPLICATIONS OF INSOMNIA AND SLEEP DEPRIVATION

Standard diagnostic frameworks classify sleep pathologies into 7 major categories: insomnias, sleep-related breathing disorders, central disorders of hypersomnolence, circadian rhythm sleep-wake disorders, parasomnias, sleep-related movement disorders, such as restless

legs syndrome, and other sleep disorders.¹⁰ From a cardiovascular and autonomic perspective, these disturbances can be broadly grouped into 2 primary domains: respiratory and nonrespiratory disorders. While SDB represents a mechanical and respiratory disruption of sleep, nonrespiratory sleep disorders also exert profound cardiovascular toxicity through overlapping autonomic pathways.^{72,73} Within this nonrespiratory domain, this review will focus specifically on insomnia and behavioral sleep deprivation.

Insomnia is increasingly recognized as an independent contributor to cardiovascular risk, beyond its association with traditional factors, such as obesity and hypertension.⁷⁴ Experienced as difficulty initiating or maintaining sleep despite adequate opportunity, insomnia is fundamentally driven by a state of 24-hour central nervous system hyperarousal that is characterized by elevated sympathetic nerve activity, increased resting heart rate, and an altered sympathovagal balance that persists

Factors Contributing to Myocardial Ischemia in SDB

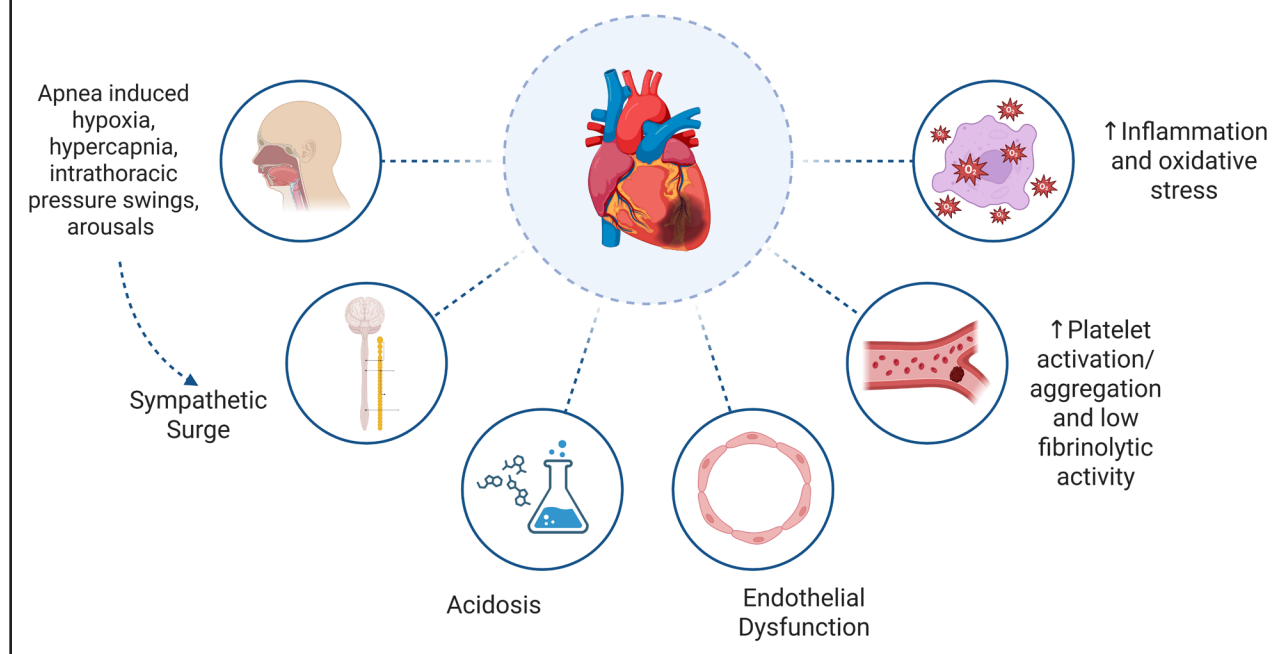


Figure 3. Key mechanisms of myocardial ischemia in sleep-disordered breathing (SDB).

Multifactorial pathways linking SDB to myocardial ischemia. The primary insults of SDB: intermittent hypoxia, hypercapnia, intrathoracic pressure swings, and recurrent arousals collectively trigger a cascade of downstream effects, including (1) massive sympathetic surges, which increase myocardial oxygen demand; (2) systemic inflammation and oxidative stress; (3) endothelial dysfunction, which impairs vascular reactivity; (4) a prothrombotic state via platelet activation; and (5) acidosis. Together, these mechanisms contribute to the ischemic burden by both increasing cardiac demand while compromising myocardial oxygen supply. Created in BioRender. Khan, S. (2026) <https://BioRender.com/huwq1ah>.

throughout both day and night.^{75,76} Crucially, this continuous sympathetic drive also blunts the normal physiological nocturnal dip in blood pressure, a well-established independent predictor of cardiovascular mortality, ultimately creating a hemodynamic milieu that promotes hypertension and vascular remodeling.^{49,77} Twenty-hour secretion of both ACTH (adrenocorticotrophic hormone) and cortisol is significantly elevated in insomnia, with the greatest increases occurring in the evening and first half of the night.⁷⁸ In addition, insomnia and sleep deprivation activate cellular and molecular processes of inflammation, including increases in NF- κ B (nuclear factor- κ B) signaling, which leads to inflammatory gene expression, production of proinflammatory cytokines, and systemic inflammation, evidenced by significant elevation in levels of IL (interleukin)-1 β , IL-6, TNF- α (tumor necrosis factor- α), and CRP (C-reactive protein).^{79–81} Mendelian randomization studies demonstrate that genetic liability to insomnia increases the risk of hypertension by 16% (odds ratio, 1.16 [95% CI, 1.13–1.18]).⁷⁴ Beyond hypertension, this systemic burden translates into significantly elevated risks for major adverse outcomes, including

myocardial infarction,⁸² heart failure,^{83,84} incident AF,^{85,86} stroke,^{74,87} and both cardiovascular and all-cause mortality.^{82,88} Notably, the combination of clinical insomnia with objective short sleep duration represents a particularly high-risk phenotype that confers additive cardiovascular risk beyond either condition alone.⁸⁹

Distinct from the hyperarousal of insomnia, behavioral sleep deprivation, defined as habitually restricted sleep duration (typically ≤ 6 hours per night), imposes its own specific cardiovascular toll. By artificially truncating the sleep period, sleep restriction disproportionately curtails the parasympathetic-dominant NREM stages, skewing the overall 24-hour autonomic balance toward sympathetic dominance.⁹⁰ Experimental sleep deprivation in normotensive adults is associated with acute rises in systolic and diastolic blood pressures of ≈ 13 and 7 mm Hg, respectively,⁹¹ and blunting of nocturnal blood pressure dipping.^{92,93} Sleep deprivation is associated with heightened inflammation,^{94,95} endothelial dysfunction,⁹⁶ and impaired insulin sensitivity and glucose metabolism.^{97,98} In experimental settings, sleep deprivation has been associated with increased appetite and weight gain, particularly

affecting visceral fat accumulation.⁹⁹ Epidemiologically, this physiological disruption drives a substantially heightened risk for incident coronary artery disease,^{100,101} myocardial infarction,¹⁰² stroke,¹⁰³ AF,^{86,104,105} and heart failure.¹⁰⁴ Ultimately, sleep duration demonstrates a well-documented U-shaped association with cardiovascular mortality, meaning that both insufficient and excessive sleep are detrimental,⁹ with objective sleep restriction below 7 hours elevating mortality risk by 29% (plus 6% per hour lost) and peaking in severity for those sleeping ≤5 hours.^{100,104}

SDB, AUTONOMIC DYSREGULATION, AND CARDIOVASCULAR RISK

SDB is a powerful driver of cardiovascular pathology, which is independently associated with a higher risk of AF, stroke, coronary artery disease, and death.^{49,106} This umbrella term encompasses a spectrum of conditions, from snoring and hypoventilation syndromes¹⁰⁷ to its primary forms that will be the focus of this article: OSA and central sleep apnea (CSA).^{107,108} OSA, the most prevalent form of SDB,¹⁰⁹ frequently coexists with and directly contributes to the development of cardiovascular diseases, acting as a key mechanistic link between sleep and cardiovascular risk.^{110,111}

Sleep disorders disrupt normal autonomic function, resulting in chronic sympathetic overactivity, impaired baroreflex regulation, and the loss of nocturnal blood pressure dipping.^{73,112} This dysregulation of autonomic balance contributes to the pathogenesis of hypertension, arrhythmias, and atherosclerosis, and is strongly associated with higher risks of myocardial infarction, heart failure, and stroke.^{113–115} The American Heart Association emphasizes that sleep disorders, particularly insomnia and OSA, not only promote traditional cardiovascular risk factors, such as obesity, hypertension, diabetes, and dyslipidemia, but also independently elevate the risk of ischemic heart disease and AF through mechanisms including systemic inflammation, endothelial dysfunction, and autonomic nervous system dysregulation.^{116,117} Understanding these interconnections between sleep, autonomic regulation, and cardiovascular health is critical for developing effective strategies to prevent and manage cardiovascular disease in populations at risk.

The multifactorial mechanisms driving this risk combine to create a powerful cyclical cascade of pathophysiological insults.⁴⁹ Each of these events triggers an acute, massive surge in sympathetic nervous system activity and blood pressure, repeated hundreds of times per night.^{46,118–121} This nightly cascade has profound chronic consequences, most importantly the development of sustained sympathetic overdrive.^{122,123} Chronic intermittent hypoxia (CIH) sensitizes the carotid chemoreflex, effectively resetting the body's autonomic baseline to a new, elevated set point. This results in a persistent, tonic

increase in sympathetic tone that endures even during daytime wakefulness, providing a direct mechanistic link to the development of chronic hypertension and heart failure.¹²⁴

The diagnostic pathway for SDB typically begins with clinical findings and patient-reported symptoms. Numerous questionnaire-based screening tools have been developed to facilitate symptom identification and identify those at high risk.^{125–127} These screening tools generally feature high sensitivity but low specificity, making them most effective for ruling out SDB rather than confirming it. Furthermore, their diagnostic accuracy is reduced in certain populations. For example, women may report atypical symptoms such as fatigue rather than classic excessive daytime sleepiness.^{128,129} Recent data suggest that artificial intelligence algorithms may serve as a good screening tool for OSA, with greater accuracy in women.¹³⁰ Similarly, patients with heart failure often do not complain of sleepiness despite experiencing significant sleep fragmentation, reducing the utility of symptom-based screening in that cohort.^{131,132}

While screening tools raise suspicion, the severity of SDB and its associated autonomic insults are formally quantified by identifying the specific respiratory events that comprise the disorder. The gold-standard method for this is an attended, in-laboratory polysomnography.¹³³ This comprehensive test simultaneously records multiple physiological signals, including electroencephalography (sleep staging), airflow, respiratory effort, and pulse oximetry. These specific channels allow for the precise identification and scoring of apneas and hypopneas. Clinically, respiratory events are defined as apneas (≥90% decrease in airflow for ≥10 seconds) and hypopneas (≥30% reduction in airflow for ≥10 seconds associated with ≥4% oxygen desaturation).^{134,135} Events are subsequently classified as either obstructive or central based on the presence or absence of respiratory effort, respectively.^{108,136}

OBSTRUCTIVE SLEEP APNEA

OSA is characterized by the repetitive collapse of the upper airway during sleep.¹³⁷ This leads to the recurrent obstructive apneas and hypopneas defined previously, where respiratory effort persists or even increases despite the cessation of airflow. The severity of OSA is quantified by the apnea-hypopnea index (AHI), which represents the total number of apneas and hypopneas per hour of sleep. A diagnosis of OSA is typically made when the AHI is ≥5 events/h in a patient with associated symptoms such as excessive daytime sleepiness or unrefreshing sleep or ≥15 events/h regardless of symptoms.¹³³ Applying these criteria, OSA is now recognized as a common, underdiagnosed disorder with a rising prevalence and substantial morbidity and mortality, particularly related to cardiometabolic and neurocognitive outcomes.^{138–140}

In the United States, the prevalence of moderate-to-severe OSA is estimated to be 17% in women and 34% in men aged 30 to 70 years, increasing dramatically with age and obesity.^{137,140} Among older adults (aged 54–93 years), prevalence can exceed 60%, with moderate-to-severe disease present in 30% to 39% of individuals, varying by race and ethnicity.¹⁴¹

From an autonomic standpoint, each obstructive event is a potent cardiovascular stressor that follows a distinct, cyclical cascade. The airway collapses, leading to a progressive fall in oxygen saturation and a rise in carbon dioxide. This escalating hypoxia and respiratory effort eventually trigger a brief, protective arousal from sleep, which reestablishes airway patency, accompanied by a massive, chemoreflex-driven surge in sympathetic nervous system activity and an acute spike in blood pressure.^{124,142} Simultaneously, the patient's repeated, forceful attempts to breathe against a closed glottis (a Mueller maneuver) create highly negative intrathoracic pressures.¹⁴³ This increases cardiac afterload and transmural pressures, adding significant mechanical stress to the

heart and great vessels.¹⁴⁴ This perfect storm of hypoxia, arousal, and intense pressure swings is repeated, potentially hundreds of times per night, driving the chronic sympathetic overdrive and endothelial dysfunction linking OSA to the development and progression of cardiovascular disease (Figure 4).^{145,146}

CENTRAL SLEEP APNEA

CSA is characterized by dysfunctional brainstem regulation of breathing with an exaggerated response to hypercapnia and sympathetic overdrive.¹⁴⁷ Under normal circumstances, the rate and depth of breathing are regulated by a negative-feedback system working to maintain carbon dioxide levels within a tight range.¹⁴⁸ Peripheral chemoreceptors present within carotid bodies and respond primarily to hypoxia but are also activated by pH, hypercapnia, hypoperfusion, hypoglycemia, and changes in temperature.^{149,150} Central chemoreceptors present within the medulla respond primarily to hypercapnia and have the greatest role in responding to hypercapnia.¹⁵¹

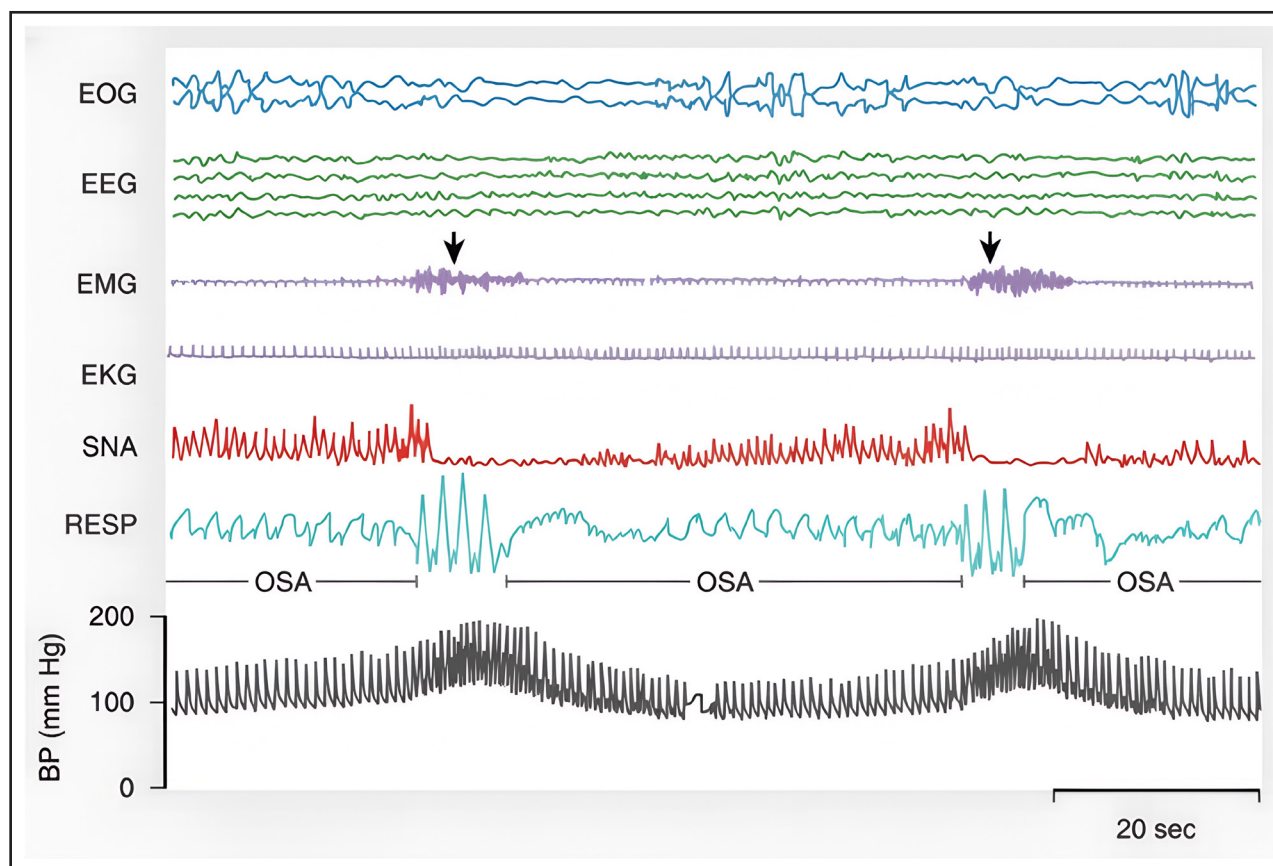


Figure 4. The autonomic and hemodynamic cascade of obstructive sleep apnea.

Original microneurography and polysomnography recordings from a patient during sleep. The channels show electrooculogram (EOG), electroencephalogram (EEG), electromyogram (EMG), ECG (EKG), muscle sympathetic nerve activity (SNA), respiration (RESP), and continuous blood pressure (BP). During the obstructive events, progressive hypoxia and hypercapnia activate peripheral and central chemoreflexes, triggering a rising cascade of SNA. This sympathetic surge drives vasoconstriction, leading to marked surges in BP. With the resumption of breathing (hyperventilation), SNA is abruptly inhibited by thoracic afferents. The black arrows indicate bursts in the EMG channel, corresponding to brief arousals from sleep, which terminate the apneic events. The EOG and EEG activities are characteristic of REM sleep, a state of autonomic volatility that can be pathologically amplified by SDB. Recreated with permission.¹¹⁹

The working theory behind the development of CSA is of dysfunctionally increased sensitivity of central chemoreceptors to carbon dioxide.¹⁵²

OSA and CSA may also both exist together. The 2 primary circumstances in which OSA and CSA may coexist are in patients with heart failure, in whom both forms may frequently present, and in cases of complex sleep apnea, where CPAP treatment may induce central apneas.¹⁵³ The underlying mechanisms behind these forms are not completely understood but are likely due to dual causation, including an anatomic substrate for airway obstruction (such as pharyngeal accumulation of edema when the patient is in the supine position), as well as physiological vulnerability to chemoreflex dysfunction with subsequent oscillations in undercorrection and overcorrection of apneas.¹⁵⁴

MECHANISTIC DYSREGULATION: THE ROLE OF INTERMITTENT HYPOXIA

SDB fundamentally transforms sleep from a period of physiological rest into a nightly cycle of profound autonomic and hemodynamic stress.^{13,109} Central to this pathology is intermittent hypoxia (IH), where the rapid oscillation between hypoxemia during apneas and subsequent reoxygenation upon resolution is a powerful driver of oxidative stress and systemic inflammation.¹⁴⁵ This oscillation also delivers a powerful, repetitive shock to the autonomic nervous system. Each significant drop in oxygen saturation is sensed by the peripheral chemoreceptors (carotid bodies), which, in turn, trigger a massive, chemoreflex-mediated surge in sympathetic nervous system activity^{142,155} as the body's desperate response to suffocation, which, in patients with severe OSA, is activated hundreds of times per night.

In healthy subjects, acute IH also shifts sympathovagal balance toward sympathetic dominance, as evidenced by increased low-frequency/high-frequency ratios in heart rate variability, elevated blood pressure, and increased catecholamine levels.¹⁵⁶ Patients with OSA demonstrate increased sympathetic activity that is not limited to acute surges following apneic events. Crucially, this heightened activity persists into the daytime despite the absence of respiratory events, likely due in part to tonic chemoreflex activation, highlighting the chronic nature and wider consequences of the disease.^{119,123,157}

Complementing this sympathetic hyperresponsiveness is a concurrent blunting of the body's primary antihypertensive compensatory mechanism. Patients with OSA exhibit downregulation of baroreflex function, evidenced by attenuated heart rate and vascular responses to baroreceptor activation.^{158,159} CPAP treatment has been shown to restore baroreflex function,¹⁶⁰ strongly supporting the understanding that CIH attenuates arterial baroreflex function. The complex role of ET-1 (endothelin-1) in this dual pathology is also of interest. OSA elicits

increases in ET-1, attenuated by CPAP.¹⁶¹ While CIH-induced upregulation of ET-1 is thought to mediate the hypersensitivity of the carotid body to hypoxia, it simultaneously depresses baroreceptor responses to increased sinus pressures, creating a vicious cycle of autonomic instability.^{162,163}

Beyond chemoreflex-driven sympathetic surges, the diving reflex represents another critical autonomic pathway activated during OSA. This primitive, mammalian reflex, responsible for conserving oxygen during submersion, is also triggered by the apneic stimulus.^{164,165} Its activation results in a powerful and simultaneous coactivation of peripheral sympathetic and cardiac vagal autonomic branches, leading to intense peripheral vasoconstriction to maintain cerebral and cardiac circulation, and potent vagal stimulation that induces bradycardia.^{166,167} In the context of OSA, this powerful vagal surge is a primary driver of nocturnal bradyarrhythmias.¹⁶⁸ This reflex-mediated slowing of the heart rate may manifest as severe sinus pauses or atrioventricular block, particularly in patients who are already susceptible to conduction abnormalities.¹⁶⁹

CIH AND CHEMOREFLEX HYPERRESPONSIVENESS

Chronic hypoxia is associated with structural changes in the carotid body, including greater vascularization, carotid body hypertrophy, and an increase in the number of oxygen-detecting cells (glomus cells).^{170,171} CIH acts as a major pathological stimulus, leading to major, long-lasting effects on the chemoreceptor system, which drives sustained sympathetic overactivity. The essential role of the carotid body in this pathology is confirmed by both clinical evidence and animal evidence.^{172,173} Subjects that have undergone cervical glomectomy (surgical excision of the carotid bodies) do not develop hypertension, even when experiencing recurrent apneas, clearly demonstrating that sustained sympathetic outflow is critically dependent on afferent signals originating from the peripheral chemoreceptors.^{174–176}

Recurrent nightly episodes of hypoxemia induce sensitization and hyperresponsiveness of the specialized oxygen-sensing peripheral chemoreceptors.¹⁷⁷ In the carotid body, CIH induces the formation of reactive oxygen species (ROS) and increased intracellular calcium levels, which drive increased expression of HIF (hypoxia-inducible factor)-1 α and decreased levels of HIF-2 α .¹⁷⁸ This dysregulated HIF activity increases HIF-1 α -dependent expression of Nox2 (NADPH oxidase 2, a pro-oxidant enzyme) and decreases HIF-2 α -dependent expression of Sod2 (superoxide dismutase 2, an antioxidant enzyme), driving persistently elevated ROS levels in the carotid body, brainstem, and adrenal medulla.¹⁷⁹ Sustained ROS production inhibits voltage-gated potassium channels in glomus cells, leading to membrane

depolarization, calcium influx, and exaggerated release of excitatory neurotransmitters even in response to modest hypoxic stimuli.¹⁸⁰ This amplified afferent drive from the peripheral chemoreceptors promotes long-term synaptic and cellular remodeling within the nucleus tractus solitarius, hypothalamic paraventricular nucleus, and rostral ventrolateral medulla, consolidating a state of chemoreflex hyperresponsiveness that supports chronic sympathetic overactivity and heightened ventilatory responses to hypoxia.^{119–121,180} During an acute apneic event, the sensitized chemoreceptors fire more intensely and rapidly than normal.¹⁸¹ This enhanced firing, even to minor drops in oxygen saturation, results in a massive, exaggerated activation of the sympathoexcitatory pathways in the brainstem, causing a disproportionate surge in sympathetic nerve activity and blood pressure upon arousal.^{180,182,183}

Supporting this theory of chemoreflex hypersensitivity, rats exposed to CIH (2 weeks) show sympathetic activation that is significantly more pronounced in response to stimuli compared with controls.¹⁸⁴ Similarly, patients with OSA demonstrate greater ventilatory responses to hypoxia compared with healthy controls.¹⁸⁵ Administration of 100% oxygen to awake patients with normoxic OSA elicits reductions in both sympathetic activity and blood pressure.¹²³ Last, despite sympathetic nerve activity increasing following apneas in controls, the magnitude of the immediate rise in sympathetic nerve activity is significantly more pronounced in patients with OSA.^{186–188} The clinical relevance is further underscored by the observation that only in patients with OSA does exposure to hyperoxia lead to a significant reduction in blood pressure.¹⁸⁹

NEUROPLASTICITY AND AUTONOMIC RESET

The transition from healthy physiological adaptation to chronic autonomic dysregulation in OSA is governed by a shift from adaptive neuroplasticity, the nervous system's ability to undergo structural and functional reorganization in response to stimuli, to maladaptive remodeling, where these adaptations become compensatory failures that sustain pathology.^{118,190} Maladaptive remodeling is characterized by progressive deterioration of synaptic function, loss of neuronal integrity, and sustained inflammatory activation that outlasts the initial hypoxic stimulus, leading to a persistent, distorted synaptic state that functionally manifests as autonomic dysregulation, altered chemoreceptor sensitivity, compromised ventilatory control, and upper-airway dysfunction.^{180,191,192} The oscillating pattern of deoxygenation and reoxygenation in CIH induces long-term neuroplasticity within the central nervous system through several interconnected mechanisms.^{178,193}

In the nucleus tractus solitarius, early IH (around 7 days after exposure) is associated with increased expression of neuroplasticity markers, including brain-derived neurotrophic factor and its receptor, TrkB (tropomyosin receptor kinase B), both of which are crucial to neuronal development, maturation, and survival.¹⁹⁰ However, extended exposure to IH (around 95 days) induces a state of neuroinflammation where brain-derived neurotrophic factor and TrkB protein expression are lower than in normoxic controls.¹⁹⁰ This transition is accompanied by glial activation and proinflammatory cytokine release (IL-1 β , IL-6, and TNF- α) in the carotid body and nucleus tractus solitarius.^{193–196} Microglial cells, the primary immune effectors in the CNS, undergo activation in response to CIH through both direct effects and indirect mechanisms involving peripheral inflammation, vagal afferent activation, and release of damage-associated molecular patterns from injured neurons,^{197,198} ultimately shifting from a surveillance phenotype that synthesizes neuroprotective factors to a proinflammatory phenotype characterized by production of neurotoxic mediators that trigger pathology including hippocampal apoptosis, impaired synaptic plasticity, and cognitive impairment.¹⁹⁹

The central serotonergic system plays a critical role as a key modulator of respiratory motor output through the induction of long-term facilitation, a form of plasticity that enhances upper-airway muscle tone following hypoxic challenges.²⁰⁰ Despite initially driving serotonin-dependent long-term facilitation of breathing, chronic upregulation of MAO-A (monoamine oxidase A) due to CIH eventually leads to serotonin depletion and oxidative stress.^{200,201} By significantly increasing MAO-A expression in the brainstem and hippocampus, CIH induces oxidative stress, NF- κ B activation, inflammation, and accelerated serotonin catabolism, evidenced by an increased 5-hydroxyindoleacetic acid to serotonin ratio, indicating a state of central serotonin depletion.²⁰² In addition, upregulated MAO-A activates the enzyme IDO-1 (indoleamine 2,3-dioxygenase-1), shunting the precursor tryptophan away from serotonin synthesis and toward the kynurenine pathway, signifying a diversion of metabolic resources that would otherwise support synaptic health and neuroplasticity.²⁰²

CIH exerts complex effects on upper-airway motor control reflecting both compensatory and maladaptive plasticity. At the level of hypoglossal motoneurons innervating the genioglossus, CIH enhances respiratory-related nerve activity and increases firing frequency and synaptic excitation of protruder and retractor motoneurons.¹⁹¹ Driven by increased intrinsic excitability, a depolarized resting membrane potential, and diminished potassium leak conductance, these physiological changes ultimately lead to maladaptive motor unit remodeling.¹⁹¹ Although motor unit amplitude increases during baseline respiration, it paradoxically falls during airway obstruction. This dysfunctional

discharge behavior points to central respiratory network alterations that may actively compromise upper-airway patency.¹⁹²

Neuroinflammation also has a substantial impact on ventilatory control. IL-1 β directly modulates respiration by reducing the amplitude of inspiratory rhythm generated by the pre-Bötzinger complex in a concentration-dependent manner.²⁰³ Furthermore, proinflammatory cytokines compromise ventilatory responses to hypoxia and hypercapnia through COX (cyclooxygenase) and nitric oxide synthase pathways, with COX inhibition eliminating IL-1 β -induced reductions in hypercapnic ventilatory response.^{204,205} To counteract IH, serotonergic neurons within the dorsal raphe and raphe magnus initially increase genioglossus corticomotor activity to prevent airway collapse. Notably, the raphe magnus loses this facilitatory capacity during chronic exposure, leaving the respiratory control system increasingly vulnerable and solely reliant on the dorsal raphe nucleus to maintain airway stability.²⁰⁶ Within the genioglossus muscle itself, CIH triggers a cascade of cellular damage, including mitochondrial dysfunction, disrupted mitophagy, and a reduction in collagen synthesis driven by the accumulation of ROS.^{207,208}

HEMODYNAMIC AND PHYSICAL INSULTS

The chemoreflex-driven sympathetic surge is further compounded by the protective arousal. The arousal itself, a brief awakening from sleep to reestablish airway patency, is a separate and distinct startle response that delivers a massive burst of sympathetic activity, which is frequently disproportionately larger than the physiological need, and which often persists beyond the arousal itself.^{209,210} Furthermore, and of particular relevance to patients with SDB, the sympathetic surge that accompanies an arousal is greatest in subjects with higher frequency or intensity arousals.^{211,212} The frequency of arousals during sleep, referred to as the arousal index, represents a significant predictor of excess sympathetic activity. Many studies have found the arousal index to be a more accurate predictor of sympathetic overactivity than the AHI.^{213,214} Even in healthy subjects (in both normoxic and hypoxic conditions), arousals are associated with an increase in sympathetic and a reduction in parasympathetic activities.²¹⁵ Beyond these direct autonomic insults, recurrent arousals are the primary driver of sleep fragmentation. This fragmentation severely degrades sleep quality, leading to significant neurocognitive consequences, such as excessive daytime sleepiness and impaired vigilance, which dramatically increase the risk for motor vehicle accidents and workplace errors.

Another major insult in OSA is the large intrathoracic pressure swings caused by repeated, forceful attempts to inspire against a closed glottis. To investigate the hemodynamic impact of these pressure swings, studies

often use the Mueller maneuver, where subjects voluntarily mimic an obstructive apnea.²¹⁶ The autonomic consequences of the Mueller maneuver are similarly striking, creating significant oscillations in cardiovascular control. The maneuver induces a steep drop in central venous pressure, along with an initial suppression of both blood pressure and sympathetic nerve activity. This dip is rapidly reversed by a dramatic late-phase increase in sympathetic nerve activity to levels far exceeding baseline, which begins to drive blood pressure back up.¹⁴³ During the maneuver, intrathoracic pressure can plummet to -30 to -40 cmH₂O, which has immediate and complex effects on the heart.²¹⁶ This profound negative pressure increases venous return to the right heart while simultaneously increasing the left ventricular transmural pressure, effectively increasing left ventricular afterload, impeding ejection, and acutely reducing stroke volume and ejection fraction.^{216,217} These changes in afterload are also reflected in the left atrium. The negative intrathoracic pressure increases transmural pressure gradients across all cardiac chambers, resulting in elevated myocardial wall stress¹⁴⁴ and acutely reducing both left atrial and left ventricular strain and filling.^{218,219} Consequently, left atrial volume acutely decreases during the maneuver, with a compensatory increase upon its release.²¹⁹ While these effects are transient, their repetition hundreds of times per night is the proposed mechanism for chronic structural damage. This repeated exposure to profound afterload and wall stress, particularly in patients with pre-existing cardiac pathologies or reduced contractile function, is thought to promote chronic left atrial enlargement and remodeling, creating a vulnerable substrate predisposing patients to AF and heart failure.^{219,220}

SYSTEMIC MEDIATORS OF INJURY: INFLAMMATION AND METABOLIC DYSREGULATION

Beyond the immediate autonomic and hemodynamic perturbations of SDB, the accompanying hypoxic burden initiates a profound cellular cascade that drives long-term cardiovascular injury. Sustained hypoxia inhibits PHD (prolyl hydroxylase domain) enzymes that require oxygen to hydroxylate HIF-1 α for VHL (von Hippel-Lindau) protein binding and degradation, leading to cytoplasmic HIF-1 α accumulation before its binding with HIF-1 β to initiate transcriptional responses for adaptation, including release of VEGF (vascular endothelial growth factor), nitric oxide synthase, and EPO (erythropoietin).^{221–223} During IH, however, rapid reoxygenation restores HIF-1 α subunit hydroxylation and subsequent degradation, leading to blunted HIF-1 α transcriptional activity. IH arising from recurring cycles of apnea and hypopnea creates a physiological milieu comparable to the ischemia-reperfusion environment seen in stroke and

myocardial infarction.^{122,224} The metabolic intermediates that accumulate during the hypoxic phase are rapidly processed upon reoxygenation, leading to a burst of ROS that quickly exhausts the cell's antioxidant defenses, driving oxidative stress and vascular injury.²²⁴ This process is further aggravated as polymorphonuclear neutrophils are activated by low oxygen tension before adhering to the vascular lining and discharging free radicals, directly damaging the endothelium.⁶¹ NF- κ B holds an essential role in the transcriptional activation of proinflammatory cytokines and adhesion molecules to produce a comprehensive inflammatory response. NF- κ B activation triggers production of the potent chemoattractant IL-8, as well as TNF- α and IL-6, both of which stimulate systemic inflammation (including hepatic production of acute-phase reactants such as CRP), insulin resistance, and metabolic dysfunction.^{225–229} The key driver, IH, also activates the NLRP3 (NLR family pyrin domain-containing 3) inflammasome protein complex that is responsible for processing the precursor of the highly potent proinflammatory cytokine IL-1 β into its mature form.^{230,231} There are a number of mechanisms linking IH to NF- κ B activation. First, IH selectively activates NF- κ B rather than adaptive HIF-directed pathways, leading to a proinflammatory response over an adaptive one.²³² Second, NF- κ B is also directly activated by mitochondrial ROS generated during reoxygenation through redox-sensitive mechanisms.²³³ Third, IH-related activation of the canonical IKK- β -dependent pathway drives phosphorylation of I κ B (inhibitor kappa B) proteins bound to inactive NF- κ B, causing their ubiquitination and proteasomal degradation, and subsequent further release of NF- κ B.²³⁴ Importantly, the magnitude of NF- κ B activation and downstream effects is directly proportional to the degree of hypoxia reached.²³⁵ Fourth, NF- κ B mediated inflammation makes use of an amplifying feedback loop, whereby TNF- α and IL-1 β are activated by NF- κ B, and TNF- α and IL-1 β activate NF- κ B, sustaining a self-perpetuating cycle of cytokine production, endothelial activation (including upregulation of VCAM-1 [vascular cell adhesion molecule-1] and ICAM-1 [intercellular adhesion molecule-1]), and progressive cardiovascular and neural end-organ injury.

The vascular endothelium represents the primary area where hypoxia-driven inflammation and oxidative stress converge. NF- κ B binding to transcriptional response elements also drives upregulation of VCAM-1 and ICAM-1, facilitating endothelial activation and leukocyte recruitment.^{236–238} Monocytes from patients with OSA bind more actively to cultured endothelial cells than those taken from controls, and CPAP therapy is associated with attenuation of this enhanced binding.²³⁹ Through leukocyte adhesion and transendothelial migration, monocytes seeded within the intima differentiate into macrophages and later into foam cells under the influence of oxidized lipids and NF- κ B-dependent signaling.^{234,240} This adhesion process

serves as the requisite first step for immune cell infiltration, effectively compromising vascular integrity and fostering an environment of chronic inflammation that leads to endothelial dysfunction. Superoxide anions generated during reoxygenation react rapidly with nitric oxide to form peroxynitrite, a highly cytotoxic oxidant.²⁴¹ ROS and NF- κ B activation concurrently uncouple eNOS (endothelial nitric oxide synthase) activity and reduce nitric oxide bioavailability.²⁴² Uncoupled eNOS switches from nitric oxide synthesis to superoxide generation, further adding to oxidative stress and the imbalance of vasoactive mediators, while favoring ET-1 and other vasoconstrictors, leading to impaired endothelium-dependent vasodilation and promotion of vasoconstriction, arterial stiffness, and microvascular rarefaction.^{243–245} Combined, these changes produce a state of endothelial dysfunction characterized by impaired vascular reactivity, heightened adhesion and coagulation factor expression, and increased susceptibility to atherothrombosis, providing a direct mechanistic bridge between IH, systemic inflammation, and downstream cardiovascular events.^{246,247}

Crucially, this inflammatory cascade is inextricably linked to the profound autonomic dysregulation characteristic of SDB, forming a bidirectional, feed-forward loop. Heightened sympathetic nerve activity and the associated surge in circulating catecholamines directly stimulate β -adrenergic receptors on immune cells, promoting the proliferation of granulocyte-macrophage progenitors and subsequent myeloid cell development, driving the rapid mobilization of proinflammatory monocytes from the bone marrow and spleen into the systemic circulation.^{248–250} Conversely, the resultant systemic inflammation also exerts powerful central autonomic effects. Circulating cytokines, particularly IL-1 β , TNF- α , and IL-6, can bypass the blood-brain barrier at circumventricular organs or signal via vagal afferents to stimulate key central autonomic control centers, including the paraventricular nucleus of the hypothalamus and the rostral ventrolateral medulla further amplifying sympathetic outflow.^{251–254} Ultimately, this synergistic activation of autonomic and inflammatory pathways propagates the acute, nocturnal stressors of IH into the daytime, creating a continuous 24-hour cycle of neurocardiovascular risk.¹²³ This sustained autonomic and immunologic activation provides the critical pathophysiological foundation driving the onset of chronic hypertension, heart failure, and arrhythmogenesis.

Beyond the autonomic and immunologic systems, this 24-hour cycle of systemic stress is further compounded by profound neuroendocrine disruption, specifically through the chronic overactivation of the hypothalamic-pituitary-adrenal axis. CIH, recurrent arousals, and large intrathoracic pressure swings all contribute to this chronic stimulation. IH, in particular, has been shown to sensitize the hypothalamic-pituitary-adrenal axis, leading to exaggerated ACTH and cortisol responses to stress and

higher baseline cortisol levels.^{255,256} Independently, the sleep fragmentation driven by arousals is associated with increased adrenal gland size, suggesting chronic stimulation, and attenuation of the normal cortisol awakening response, indicating impaired circadian regulation.^{257–259} Similarly, in the rat model, CIH is associated with markedly increased catecholamine secretion from the adrenal medulla.²⁶⁰ In these studies, rat adrenal medullary chromaffin cells demonstrated facilitated calcium ion influx and mobilization of intracellular calcium stores through ryanodine receptor activation, as well as a greater number of secretory vesicles. The large intrathoracic pressure swings further amplify this neuroendocrine response by increasing sympathetic outflow.^{137,261} Collectively, these mechanisms result in elevated and disordered ACTH and cortisol secretion and impaired feedback regulation, with profound downstream effects on metabolic, cardiovascular, and inflammatory pathways. Importantly, this link is reinforced by evidence that CPAP therapy can help normalize hypothalamic-pituitary-adrenal axis activity, reducing cortisol levels and restoring circadian patterns.^{256,259,261}

SPECIFIC CARDIOVASCULAR CONSEQUENCES

Sleep, SDB, and Chronic Cardiovascular Disease

The profound autonomic insults of SDB and the resulting 24-hour sympathetic overactivity are likely the primary drivers for the development of chronic cardiovascular sequelae.^{262–264} Here, we will focus on specific cardiovascular consequences of SDB and the associated autonomic dysfunction.

Hypertension

The relationship between SDB and systemic hypertension is one of the most robust and well-established in cardiovascular medicine. Today, OSA is recognized as the most common secondary and, critically, potentially treatable cause of hypertension.²⁶⁵ The first manifestation of this disruption is the attenuation of the nocturnal dip in the blood pressure profile.²⁶⁶ Recurrent surges of sympathetic nerve activity, which are secondary to apneic events and arousals in OSA, blunt or abolish the $\approx 10\%$ to 20% reduction in mean arterial pressure that typically occurs in healthy individuals.^{267–269} Absent nocturnal dipping is itself a powerful independent predictor of cardiovascular mortality.^{114,270} The increased blood pressure in patients with OSA is not limited to sleep. The increased sympathetic nerve activity that is associated with OSA persists in wakeful hours. Together, blunted nocturnal dipping paired with persistent 24-hour sympathetic overactivity provides the mechanistic basis for chronic,

sustained daytime hypertension.^{124,271} A growing body of evidence has identified the arousal index as a powerful, dose-dependent predictor of hypertension.¹⁵⁸ In patients with OSA, for example, a 10-unit increase in the arousal index is associated with a 10% greater likelihood of prevalent hypertension.^{272,273} This dose-dependent relationship is also observed, and is particularly strong, for arousals occurring during REM sleep, which correlate with both the likelihood and severity of hypertension.²⁷³ This is not a finding confined to SDB, as the association between a high arousal index and hypertension has also been clearly demonstrated in large, general population cohorts.^{8,269}

Substantial heterogeneity exists between outcomes of studies investigating the effects of CPAP therapy on hypertension in individuals with OSA. Most studies have demonstrated modest reductions in systolic and diastolic blood pressures at best (in the order of 2–5 mm Hg).^{274–276} However, this modest net effect is in the context of significant heterogeneity across studies, which vary considerably in their design and patient characteristics. This variability has been instructive, revealing that the antihypertensive effect of CPAP is not uniform. Instead, the benefit appears to be concentrated in certain high-risk phenotypes. Specifically, greater reductions in blood pressure are consistently observed in patients with higher CPAP adherence (often defined as ≥ 4 hours per night), resistant hypertension, absent nocturnal dip, or higher baseline blood pressure.^{274,277,278} Last, this complexity is further highlighted by the finding that standard markers of OSA, such as the AHI, are unreliable in predicting which patients will experience a significant blood pressure response to CPAP therapy, underscoring the conflicting results in the literature and indicating a clear need for further research to better identify treatment responders.^{279,280}

Coronary Artery Disease and Myocardial Infarction

There is a bidirectional relationship between SDB, particularly OSA, and coronary artery disease, whereby OSA is associated with a ≈ 2 -fold increase in the risk of cardiovascular events,²⁸¹ and $\approx 40\%$ of patients presenting with ST-elevation myocardial infarction demonstrate undiagnosed moderate-severe OSA.²⁸² Patients admitted with myocardial infarction and subsequently diagnosed with comorbid OSA are more likely to have had the event during nighttime hours, likely driven by acute overnight hypoxia, adrenergic activation, inflammation, and hemodynamic stress induced by obstructive apneas.²⁸³ Patients with OSA show clear evidence of endothelial dysfunction, primarily through oxidative stress and systemic inflammation induced by CIH.^{238,242,284–286} Rapid reoxygenation following apneic events induces superoxide generation that aggressively scavenges protective nitric oxide,

converting it into the highly cytotoxic peroxynitrite.²³⁸ This oxidative burden, paired with concurrent NF- κ B activation, leads to reduced eNOS activity and hence depletion of nitric oxide availability, further superoxide generation, and upregulation of vasoconstrictors, such as ET-1, ultimately resulting in compromised vascular reactivity, and premature atherosclerosis and coronary artery calcification.^{243,284,287–289} Furthermore, the disorder contributes to coronary plaque instability and increased plaque vulnerability, with retrospective angiographic analyses confirming that the overall severity of coronary artery disease is significantly greater in patients with moderate-to-severe OSA, entirely independent of other traditional cardiovascular risk factors.^{290–292} Supporting this, a study utilizing the SYNTAX score measure of coronary artery disease complexity reported that moderate-to-severe OSA significantly correlated with greater disease complexity (partial correlation=0.24; $P=0.039$), even after comprehensive adjustment for age, sex, obesity, hypertension, hyperlipidemia, diabetes, smoking, and family history.²⁹² Ultimately, OSA acts as both a chronic accelerator of atherosclerosis and an acute physiological trigger for ischemic events, solidifying its role as a profound catalyst of cardiovascular risk.

Heart Failure

The relationship between SDB and heart failure may be considered a vicious cycle with each disease potentiating the other.^{293,294} This relationship is particularly complex in the case of CSI, which may arise as a consequence of heart failure that subsequently acts to worsen cardiac function. CSA, which often presents as Cheyne-Stokes respiration, is itself not a cause of heart failure but may develop as a direct result of the pathophysiology associated with heart failure.¹¹ Pulmonary congestion in patients with heart failure may lead to activation of the Hering-Breuer reflex, where pulmonary stretch receptors, which normally work to protect the lungs from overexpansion by reducing respiratory drive, are strongly activated, leading to central apneas.^{295,296} Some patients using CPAP therapy for treatment of OSA can also subsequently develop what is termed treatment-emergent CSA, particularly if the therapy for OSA is associated with excessive lung expansion, which then induces central apneas.^{297,298} Furthermore, as described earlier, pulmonary congestion in patients with heart failure, combined with prolonged circulation time between the pulmonary vasculature and central chemoreceptors, creates a state of high loop gain, or respiratory instability, which ultimately triggers delayed and exaggerated responses to changing levels of carbon dioxide.^{299,300}

Heart failure is itself already considered a state of chronic sympathetic activation due to compensatory but ultimately maladaptive mechanisms. In the presence of comorbid SDB, additional surges of sympathetic

activation join this already-high baseline.^{117,301} These effects directly counteract the intended therapeutic effects of β -blockade, increasing the patient's heart rate, myocardial oxygen demand, and systemic afterload.³⁰² In patients with heart failure and OSA, the mechanical stress that exists as a consequence of large, intrathoracic pressure swings leads to further increases in left ventricular afterload.^{117,294,301} In conditions such as ischemic cardiomyopathy, or others where myocardial oxygen demand is too great, recurrent IH associated with SDB also imposes a significant oxidative and ischemic burden on the already-compromised myocardium, potentially promoting inflammation and fibrosis.^{294,303} Studies assessing the impact of therapy for SDB in patients with heart failure have consistently demonstrated associated improvements in sleep and cardiac function; however, these findings have not translated into improved mortality.^{304,305}

Atrial Fibrillation

Observational and cohort studies have consistently demonstrated a close relationship between SDB and arrhythmias, particularly AF. OSA is considered an important, independent risk factor for AF, where more severe cases of OSA are associated with a greater incidence of AF, a higher burden of AF, and increased recurrence of AF following rhythm-controlling interventions such as cardioversion or left atrial ablation. While estimates of the prevalence of AF among individuals with SDB vary considerably, from 21% to 74%, depending on the population assessed, they are consistently higher than in the non-SDB population (3%–49%).^{306–309} Conversely, the prevalence of SDB is also significantly higher in patients with a greater burden of AF.³¹⁰ The association between high-frequency arousals from sleep and an approximately one-third greater risk of incident AF has been highlighted.³¹¹ Furthermore, the likelihood of incident AF in cases of moderate-to-severe OSA increases to nearly 5-fold higher than in nonapneics.³¹¹ Some of these associations may be explained by the risk factors that OSA and AF share, including obesity, older age, hypertension, male sex, and smoking, for example.^{312–315}

Repeated episodes of inspiration against airway obstruction in OSA, followed by hypoxia, pulmonary stretch inhibition, and arousals, ultimately lead to a net increase in sympathetic activity to peripheral blood vessels. Simultaneously, the profound vagal surges driven by the diving reflex induce marked bradycardia. This resulting autonomic conflict, a sudden sympathetic burst paired with intense vagal tone, increases the likelihood of triggering and maintaining AF. This is supported by the finding of attenuation of inducibility of AF in cases of OSA following interventional- or pharmacological-neurohormonal blockade.³¹⁶ The repeated swings in intrathoracic pressure and resultant

increased afterload and wall stress may also lead to atrial remodeling that further increases the likelihood of AF (Figure 5).^{317,318} Severe OSA is associated with left atrial size and area that are increased by ≈ 3 mm and 1.7 cm², respectively.^{319–321}

The arrhythmogenic risk from SDB is not confined to sleep but extends into daytime hours. For example, nights with more severe SDB are associated with a 2-fold increase in the risk of episodes of AF during the following day, supporting a clear temporal and potentially mechanistic link, suggesting that the nocturnal insults of SDB induce a lasting carry-over effect on daytime cardiac electrical stability.³²² Invariably, a number of studies have demonstrated a correlation between the presence of SDB and an increased incidence of cardioembolic stroke.³²³ Similarly, there is a significantly (up to 5-fold) greater prevalence of AF among patients with a history of OSA and stroke.³²⁴ The greater prevalence of AF in OSA is likely multifactorial, resulting from the combined effects of autonomic dysregulation, chronic mechanical stress on the left atrium, and the subsequent development of pathological atrial remodeling. This remodeling

includes atrial fibrosis, altered conduction, and changes in refractoriness.³²⁵ Animal models show that this fibrosis can be induced by OSA independent of AF,³²⁶ and human studies note that it disproportionately affects the left atrium, a finding that aligns with biatrial voltage distribution studies and supports the role of atrial fibrosis in AF pathogenesis in OSA.^{327,328}

As is the case in hypertension, reports of the efficacy of CPAP therapy in reducing the burden of AF vary considerably. Observational data support a role for CPAP in lowering the risk of AF recurrence, particularly in the postcardioversion or postablation population,^{311,329} as well as reducing the progression from paroxysmal to permanent AF.^{117,329–331} However, despite this strong mechanistic rationale, clinical trial evidence for using CPAP to treat established AF remains ambiguous. Several randomized controlled trials investigating AF recurrence have yielded conflicting results, with most, though relatively small and potentially statistically underpowered, demonstrating no significant net benefit.^{311,332–334} This discrepancy may be explained by 2 key factors. First, trial results are often confounded by highly variable patient

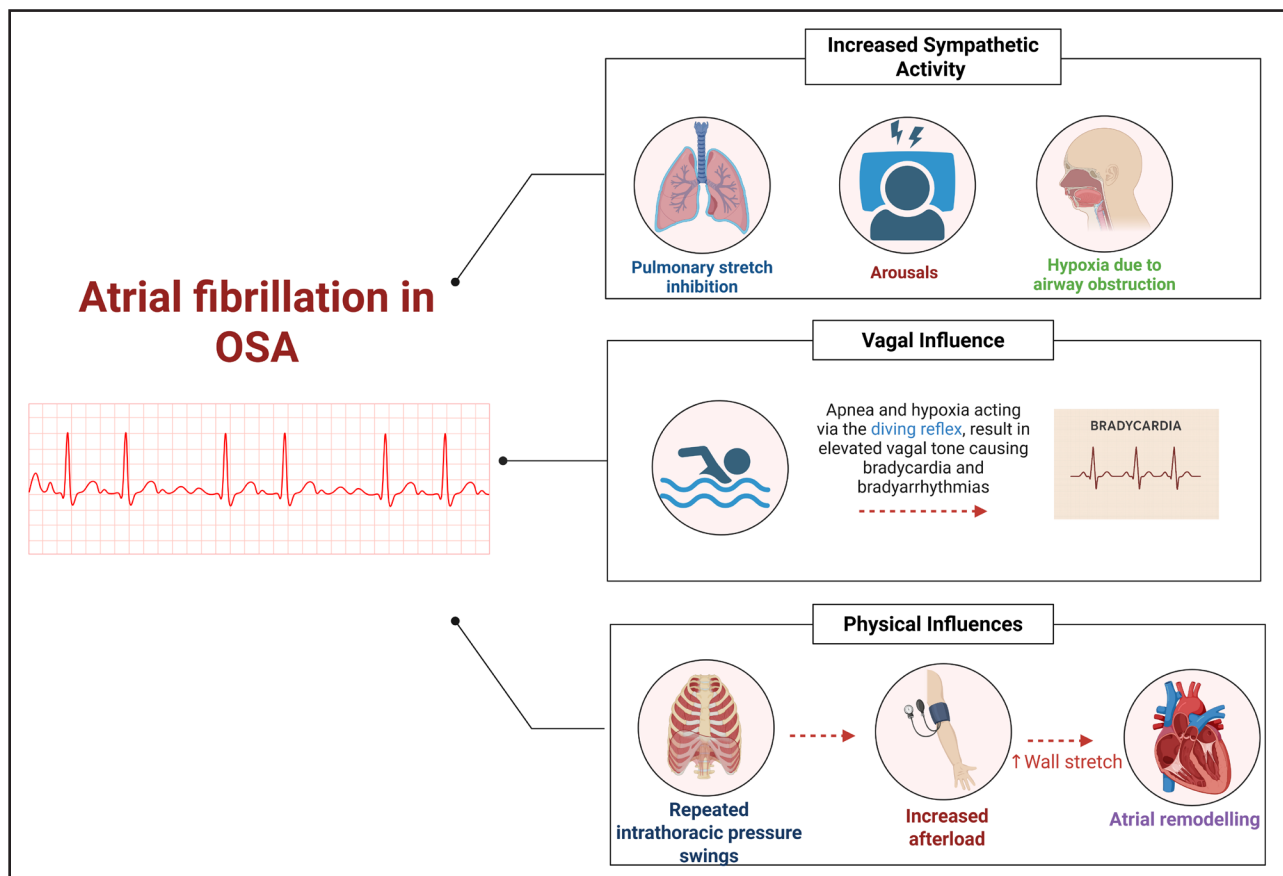


Figure 5. Multifactorial mechanisms of atrial fibrillation in obstructive sleep apnea.

This figure illustrates the combination of autonomic and physical insults by which obstructive sleep apnea (OSA) promotes atrial fibrillation (AF). Recurrent arousals and intermittent hypoxia trigger intense surges in sympathetic activity, while the diving reflex simultaneously activates a powerful vagal influence, causing bradycardia. This proarrhythmic autonomic conflict, combined with the physical stress of large intrathoracic pressure swings, increases cardiac afterload and wall stretch. Chronically, these insults are thought to drive pathological atrial remodeling and fibrosis, creating a vulnerable substrate for AF. Created in BioRender. Khan, S. (2026) <https://BioRender.com/qpwoqu6>.

adherence to CPAP. Second, these neutral findings may highlight an important distinction between treating the trigger and reversing the substrate for AF. CPAP therapy is highly effective at eliminating the acute triggers of SDB, such as IH and sympathetic surges. However, in patients with long-standing disease, their atria may have already undergone significant, irreversible structural remodeling, such as fibrosis. Therefore, initiating CPAP at this late stage may be insufficient to reverse an already-established fibrotic substrate, suggesting that its benefit may be greatest in primary prevention or in patients with early or only paroxysmal AF.

STROKE

The relationship between SDB and stroke is synergistic and reciprocal. While acute cerebrovascular events disrupt respiratory drive and precipitate SDB, preexisting OSA serves as a potent catalyst for ischemic stroke through the convergence of hemodynamic, structural, and arrhythmogenic mechanisms.^{335–337} Profound negative intrathoracic pressure swings that are characteristic of obstructive events stimulate paradoxical right-to-left shunting via a patent foramen ovale, while concurrent IH, oxidative stress, upregulated vascular inflammation, and sympathetic surges induce endothelial dysfunction, and accelerated carotid atherosclerosis and plaque vulnerability.^{156,223,338–342} In addition, simultaneous promotion of AF, hypertension, and hypercoagulability further increases embolic risk.^{275,343,344} An AHI ≥ 20 (moderate-severe OSA) has been independently associated with a doubling of stroke risk in women and a 4-fold increase in men.³⁴⁵ Similarly, the prevalence of SDB among stroke survivors (71%) is approximately 4-fold greater than in the general population.³⁴⁶

SDB impairs cerebral perfusion reserve and cerebral autoregulation, increasing the likelihood of ischemia.^{347,348} This hemodynamic compromise provides a mechanistic explanation for the high prevalence of subclinical cerebrovascular disease in this population, manifesting as a greater burden of lacunar infarcts and white matter hyperintensities.¹¹⁶ Following a stroke, comorbid SDB is associated with worse functional and cognitive recovery, as well as a higher risk of recurrent stroke and death, as the combination of ongoing recurrent hypoxemia, sleep fragmentation, and sympathetic activation blunts neuroplasticity and reduces the efficacy of rehabilitation.³⁴⁹ Addressing this risk remains a complex therapeutic challenge. Similar to other cardiovascular outcomes, meta-analyses of CPAP efficacy for stroke recurrence are heavily confounded by poor device tolerance, often masking definitive benefits.^{350–352} However, robust observational cohorts and propensity-matched, adherence-based analyses demonstrate a significant protective effect for compliant CPAP users.^{353–355}

SDB IN HIGH-RISK STRUCTURAL HEART DISEASE

The relationship between SDB and structural heart disease is particularly relevant in specific, high-risk populations. In patients with hypertrophic cardiomyopathy, for example, SDB has been identified as a highly prevalent and frequently underdiagnosed comorbidity.^{356–360} A prospective study using criterion-standard polysomnography in a cohort of patients with hypertrophic cardiomyopathy without a prior diagnosis of SDB found that previously undiagnosed SDB was present in 59% of participants.³⁵⁸ While predominantly obstructive, the prevalence of CSA was also notable. SDB was linked to elevated baseline troponin-T levels and, critically, a greater overnight increase in troponin-T, suggesting that SDB may contribute to ongoing, subclinical myocardial injury in this population.³⁶¹ Notably, the subgroup with CSA demonstrated the most adverse structural changes, including markedly higher left ventricular mass than those with OSA. In addition, traditional screening symptoms, such as self-reported snoring and daytime sleepiness, were not predictive of SDB, highlighting the limitations of standard screening tools in this population and the high potential for underdiagnosis.³⁵⁸

The presence of SDB was associated with a significantly more adverse cardiac profile, including greater left ventricular mass, more significant diastolic dysfunction (higher E/e' ratios), and worse functional status (New York Heart Association class).^{358,362} Importantly, because these specific echocardiographic parameters, progressive hypertrophy and impaired diastolic compliance, manifest before the onset of irreversible, end-stage heart failure, they may serve as highly actionable proximal clinical end points.^{363,364} Serial assessment of left ventricular mass and E/e' ratios could, therefore, provide a noninvasive window to detect early SDB-driven disease progression and gauge the efficacy of targeted sleep interventions in these vulnerable patients.^{365,366}

This vulnerability extends to the broader population of patients with adult congenital heart disease, which has a unique and complex predisposition to SDB. Many adult patients with congenital heart disease already exhibit significant baseline autonomic dysfunction as a consequence of long-term hemodynamic stress or iatrogenic surgical denervation, characterized by a neurohormonal profile similar to chronic heart failure with elevated levels of norepinephrine.^{367–369} The autonomic instability of SDB would further complicate this neural circulatory milieu. Patients with cyanotic defects often have blunted chemosensitivity from chronic hypoxemia, which disrupts respiratory control and predisposes them to CSA.^{370,371} Furthermore, the large negative intrathoracic pressure swings during obstructive events can acutely increase right-to-left intracardiac shunting in those with residual defects, further impairing systemic oxygenation.³⁷²

Worsening nocturnal hypoxemia may also heighten thrombotic risk in cyanotic patients by driving excessive erythropoiesis.^{373,374} This deleterious feedback loop, where congenital anomalies predispose patients to SDB, and SDB, in turn, exacerbates cardiac dysfunction, likely creates a cycle of progressive cardiovascular morbidity. As this uniquely vulnerable population continues to grow rapidly, the dearth of understanding of SDB's true prevalence, impact, and optimal treatment strategy in adult congenital heart disease highlights a critical area for future investigation.

SLEEP AS A TRIGGER FOR ARRHYTHMIC EVENTS

The cardiovascular risks of sleep are not limited to the chronic, maladaptive changes driven by SDB.³⁵⁸ Sleep itself is a period of profound autonomic fluctuation that may induce arrhythmia in susceptible individuals, a risk that is pathologically magnified in the presence of SDB. This is particularly evident in patients with inherited channelopathies such as long-QT syndrome (LQTS) and Brugada syndrome. In these conditions, the underlying electrical instability (ie, arrhythmogenic substrate) may be triggered by the normal physiological shifts between NREM's parasympathetic dominance and REM's sympathetic activation, leading to life-threatening events. The following section will briefly explore how these autonomic triggers interact with specific, preexisting vulnerable substrates, such as those found in channelopathies and epilepsy.

RIGHT VENTRICULAR OUTFLOW TRACT VENTRICULAR TACHYCARDIA

Representing the most common form of idiopathic ventricular tachycardia (VT), right ventricular outflow tract VT (RVOT-VT) accounts for around 70% of all cases of idiopathic VT. The vast majority of cases occur without evidence of underlying structural cardiac disease^{375,376} and present with monomorphic VT.^{375–377} While generally considered to have a benign prognosis, RVOT-VT can rarely degenerate into polymorphic VT or ventricular fibrillation, especially in the presence of frequent premature ventricular contractions.³⁷⁸ The defining pathophysiological feature of RVOT-VT is its mechanism of initiation. It is a catecholamine-sensitive, adrenergically driven arrhythmia. Myocytes in the RVOT are uniquely sensitive to the effects of adenosine and catecholamines, with the arrhythmia being mediated by cAMP-dependent triggered activity.^{375,377} This specific pathophysiology explains why RVOT-VT episodes are commonly triggered by situations that increase sympathetic tone, such as physical exercise or acute emotional stress.^{375,379} Consequently,

SDB emerges as a critical, and often overlooked, risk factor, as it induces a state of profound sympathetic overactivity,³⁸⁰ which persists even during daytime wakefulness,³⁸¹ thereby providing the requisite adrenergic trigger required to initiate episodes of RVOT-VT.

Furthermore, OSA may contribute to this arrhythmia through a second, structural mechanism. Beyond providing the acute trigger, the chronic insults of OSA (such as IH, sleep fragmentation, and large intrathoracic pressure swings) induce changes in cardiac structure and function. This includes right ventricular remodeling, which may, over time, create a vulnerable arrhythmogenic substrate in the right ventricular outflow tract, further predisposing the patient to arrhythmias.^{382,383} Therefore, in this population, OSA may act as a crucial pathological driver, providing both the acute trigger and adding to the chronic substrate.

BRUGADA SYNDROME

In contrast to the adrenergically driven mechanism of RVOT-VT, Brugada syndrome is an autosomal dominant channelopathy³⁸⁴ that predisposes individuals to syncope and sudden cardiac death³⁸⁵ and is primarily directed by a different autonomic trigger, high vagal tone.^{386,387} The underlying pathophysiology is linked to adrenergic dysfunction, with an overall reduction in norepinephrine levels and β -adrenoceptor stimulation.³⁸⁸ This relative lack of protective sympathetic input, when combined with a surge in vagal activity, unmasks the underlying electrical defect. This autonomic imbalance is what provokes the characteristic coved ST-segment elevation and subsequent arrhythmogenesis,³⁸⁸ providing a mechanistic basis for the higher prevalence of arrhythmic events during sleep in Brugada syndrome.^{389,390} Up to 93% of episodes of ventricular fibrillation are detected at night, with $\approx 87\%$ detected during sleep, particularly between the hours of midnight and 6 AM.³⁹⁰

Sleep, particularly NREM sleep, is a state of parasympathetic dominance. However, the autonomic instability of REM sleep may also influence risk. Patients with Brugada syndrome exhibit abnormal heart rate variability and altered cardiac autonomic control, suggesting that they cannot properly manage the profound physiological shifts between sleep stages.³⁸⁹ SDB is notably more prevalent in patients with Brugada syndrome, a finding that holds true even in the absence of obesity,³⁹¹ and may potentially be an underappreciated risk amplifier in this population. The recurrent apneas and hypoxias of SDB could act to further disrupt autonomic regulation, particularly by amplifying parasympathetic dominance during the hypoxic, initial phase of the apneic event, conceivably increasing the risk of sleep-related ST-segment changes and malignant arrhythmias in this highly vulnerable cohort.³⁹¹

LONG-QT SYNDROME

LQTS is an inherited channelopathy characterized by delayed ventricular repolarization that is associated with a prolonged QT interval on the ECG and a high risk for life-threatening arrhythmias, primarily Torsades de Pointes. Unlike the more uniform triggers of RVOT-VT or Brugada syndrome, the specific autonomic states that provoke arrhythmias in LQTS are highly dependent on the patient's underlying genotype and, subsequently, ion-channel defect. The risk profiles for the most common genotypes are markedly different. LQT1 (long QT syndrome type 1) occurs due to mutations in the *KCNQ1* gene, leading to abnormal function of voltage-gated potassium channels that are critical for repolarization, impairing the heart's ability to shorten the QT interval during times of adrenergic stimulation, offering an explanation for why most events occur during sympathetic surges, including exercise or periods of emotional stress.³⁹² Consequently, β -blockade and the relative vagal dominance of sleep are often protective for this group.

Patients with LQT3 (long QT syndrome type 3; caused by mutations in the *SCN5A* gene), in contrast, are at the highest risk of arrhythmia during periods of rest or sleep. Events are typically triggered by bradycardia, making the high vagal tone of NREM sleep a period of peak vulnerability³⁹² and explaining the relative inefficiency of β -blockade in this population.³⁹³ Conversely, patients with LQT2 (long QT syndrome type 2), due to mutations in the *KCNH2* gene, represent a third pattern, where events are frequently precipitated by sudden, abrupt sympathetic surges from a resting state, such as when startled by a loud noise, such as an alarm clock or telephone.³⁹⁴ This genetic variability makes sleep a uniquely dangerous period. The autonomic lability of REM sleep and, most importantly, the abrupt sympathetic surges associated with arousals create a clash between a high underlying vagal tone (sleep) and a sudden sympathetic burst (arousal), acting a potent trigger for Torsades de Pointes, particularly in LQT2.³⁹⁴ This interaction is highly specific; for instance, women with LQT2 (but not men, nor patients with LQT1) exhibit marked QTc prolongation and increased arrhythmic risk specifically during REM sleep, indicating a complex sex- and gene-specific interaction.³⁹⁵

The recurrent hypoxic events and sympathetic-driven arousals in SDB provide the precise triggers to which patients with LQT2 and LQT3 are vulnerable. The severity of OSA has also been shown to correlate directly with the degree of QT-prolongation and overall arrhythmic risk in patients with LQTS.^{396,397} Abnormal autonomic control, including impaired heart rate and QT variability, is a key determinant of adverse outcomes in this population.^{398,399} Therefore, arrhythmic risk in LQTS is determined by the complex relationship between 3 primary factors: the patient's underlying genotype, the specific sleep stage, and the overall autonomic state.

SUDDEN UNEXPECTED DEATH IN EPILEPSY

Sudden unexpected death in epilepsy represents a catastrophic, sleep-associated outcome^{400,401} of the disease, with key risk factors including generalized tonic-clonic seizures, refractory epilepsy, and prone posture.⁴⁰² Seizures arising from NREM sleep⁴⁰³ are often longer and more severe^{404,405} and have a greater detrimental impact on cardiac activity,⁴⁰⁶ thereby increasing the risk of sudden unexpected death in epilepsy.⁴⁰¹

The mechanism of death is via catastrophic failure of cardiorespiratory function, typically in the postictal period. This profound autonomic instability, characterized by a failure of brainstem regulation,^{402,407} leads to severe central apnea, catastrophic hypotension, and fatal hypoxia.⁴⁰⁸ SDB may be a critical risk amplifier, and patients with epilepsy have a high prevalence of SDB.⁴⁰⁹ This creates a dangerous milieu of heightened risk of sudden cardiac death that predominantly occurs at night.^{410,411} The combination of autonomic dysfunction and chronic hypoxia associated with SDB may prime the cardiorespiratory system, lowering the threshold for postictal collapse.⁴¹² While the precise terminal event is unclear, a primary cardiogenic mechanism is strongly suspected,⁴¹³ possibly linked to pathogenic variants in arrhythmia-related genes, similar to other channelopathies.⁴¹⁴

TRANSLATING MECHANISMS TO PROXIMAL CLINICAL END POINTS

A central challenge in sleep medicine is identifying the threshold at which the cyclical insults of SDB progress from transient, compensatory physiological stress to irreversible cardiovascular injury, such as atrial fibrosis, diastolic dysfunction, and decompensated heart failure.^{109,415} To enhance clinical translation, the mechanistic pathways described above must be anchored to measurable, proximal clinical end points that identify vulnerable individuals before overt cardiovascular disease emerges.¹¹⁷

Systemic inflammation and hemodynamic load can be quantified using routinely available serological markers. hs-CRP (high-sensitivity CRP) serves as a proxy for low-grade inflammatory activation, BNP (B-type natriuretic peptide) reflects dynamic alterations in chamber wall stress and neurohormonal burden, and high-sensitivity troponin provides a sensitive measure for subclinical myocardial injury.^{416–418} Similarly, recurrent mechanical strain due to exaggerated negative intrathoracic pressure manifests structurally as progressive atrial remodeling, quantifiable through echocardiographic markers, such as increased left atrial volume index and impaired global longitudinal strain, which often precede symptomatic heart failure.^{419,420} In parallel, the evolving arrhythmogenic substrate, shaped by autonomic imbalance and fibrotic remodeling, can be interrogated noninvasively using

electrical phenotyping techniques, such as the signal-averaged ECG, to detect ventricular late potentials or prolonged P-wave indices.⁴²¹ Integrating these serological, structural, and electrical measures into standard assessment frameworks could allow clinicians to stratify cardiovascular risk in SDB, identify preclinical dysfunction, and intervene before the substrate becomes irreversible.

CONCLUSIONS

This review establishes that sleep is not simply a passive, restorative state but an active, complex period of profound autonomic fluctuation, even in normal circumstances. The cyclical shifts between the parasympathetic dominance of NREM sleep and the sympathetic volatility during REM sleep create a baseline of cardiovascular vulnerability. SDB, OSA in particular, exacerbates and pathologically amplifies this volatility, transforming sleep into a nightly cascade of severe and repeated autonomic and hemodynamic insults. This relentless nocturnal stress provides a clear, mechanistic link for OSA as a primary driver of chronic cardiovascular diseases, including hypertension, AF, and heart failure. Furthermore, as demonstrated in channelopathies, even the normal autonomic shifts during sleep can trigger fatal arrhythmias in patients with a pre-existing vulnerable substrate, a risk that may be heightened further in the presence of comorbid SDB.

While these mechanistic links are now well-established, a crucial translational gap remains. Moving forward, the critical next step for the clinical and electrophysiology communities is translating these profound mechanistic insights into actionable preventative practice. Several key unanswered questions remain, specifically regarding the best signals to identify the tipping point between reversible autonomic stress and permanent cardiovascular injury, and whether early treatment of SDB can successfully halt the progression of the arrhythmogenic substrate. Future investigation must prioritize the clinical validation of proximal end points, shifting the focus toward early, measurable targets rather than distal disease end points. Bridging this divide between mechanistic insight and clinical detection is paramount for advancing precision cardiovascular medicine, allowing for the accurate stratification of high-risk phenotypes and the deployment of targeted, disease-modifying interventions well before the pathological cascade becomes irreversible.

ARTICLE INFORMATION

Received November 18, 2025; accepted May 28, 2026.

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Acknowledgments

Graphic abstract created in BioRender. KARIM, S. (2026) <https://www.biorender.com/0872cpm>.

Sources of Funding

V.K. Somers is supported by the National Institutes of Health grants HL65176 and HL160619.

Disclosures

V.K. Somers has served as a consultant for Lilly, Jazz Pharmaceuticals, ApniMed, GEM Health, and iRhythm. The other authors have seen and approved the manuscript and report no conflicts.

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